

Supplement to Auditing Search-Budget Portability Across Graph-ANNS Rebuilds

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A Guarantees and Their Experimental Contact

The main paper contains all central findings, including the stricter certification result and completed acquisition costs. This supplement expands the theoretical reasoning and provides numeric details needed to inspect the same findings.

A.1 Observation-Limited Loss

Let two environments have observable laws P_0^O, P_1^O , disjoint acceptable-decision sets D_0, D_1 , and losses satisfying $L_i(a) \geq \lambda \mathbf{1}\{a \notin D_i\}$. An action may encode execution, fallback, or abstention. From any common randomized decision rule $A(O)$, form the test that returns 0 exactly when $A \in D_0$. Under environment 0, a test error implies $A \notin D_0$; under environment 1, it implies $A \in D_0 \subseteq D_1^c$. Therefore

$$\begin{aligned} \mathbb{E}_0 L_0 + \mathbb{E}_1 L_1 &\geq \lambda \{P_0(\text{test error}) + P_1(\text{test error})\} \\ &\geq \lambda(1 - \text{TV}(P_0^O, P_1^O)). \end{aligned}$$

Taking half the sum gives the main bound. This is the classical two-point testing argument, not a new general inequality. Including the query jointly in O is essential when it is random. An efficient classifier lower-bounds distinguishability; it does not provide the upper bound on total variation needed for a positive loss lower bound. The empirical transfer rates are not numerical estimates of this expression.

A.2 A Conditional Margin Result for Selection

For a finite policy family, let r_j and c_j be risk and expected cost. Suppose $\max_j |\hat{r}_j - r_j| \leq \varepsilon_R$ and $\max_j |\hat{c}_j - c_j| \leq \varepsilon_C$. Retain $S = \{j : \hat{r}_j + \varepsilon_R \leq \delta\}$ and minimize estimated cost within S . Every retained policy is safe because $r_j \leq \hat{r}_j + \varepsilon_R \leq \delta$. If j_* minimizes true cost among policies with $r_j \leq \delta - 2\varepsilon_R$, then $j_* \in S$ and

$$c_j \leq \hat{c}_j + \varepsilon_C \leq \hat{c}_{j_*} + \varepsilon_C \leq c_{j_*} + 2\varepsilon_C.$$

The comparison is against the margin-feasible family, not all safe policies. It requires a justified uniform estimation event and excludes the subsequent cost of rejected-policy fallback. TCP's implemented selector takes the first qualifying shift, not this empirical-cost argmin; this result therefore explains a sufficient selection condition without being claimed as an implemented TCP performance guarantee.

A.3 Build-Rank Coverage

For a fixed query, assume m source labels and one target label are exchangeable, allowing $+\infty$. Choose source order statistic $r = \lceil (1-\eta)(m+1) \rceil$ when it exists and is finite, and otherwise abstain. Break ties with independent continuous variables. The augmented target rank is uniform among $m+1$ positions. A target label strictly above the chosen source value must have augmented rank greater than

r , hence

$$\Pr\{a < \infty, B_t > a\} \leq \frac{m+1-r}{m+1} \leq \eta.$$

If labels are stable-tail budgets, $B_t \leq a$ implies execution success on the finite grid. First-passing labels lack that implication without an additional monotonicity bridge. The guarantee is marginal over builds for each fixed query, not conditional on one target or simultaneous over queries. Nine source labels cannot give a nonvacuous 5% rank guarantee; nineteen are necessary at that level. The refresh replay uses nine first-passing histories from an old snapshot, so it does not instantiate these assumptions.

A.4 Fixed-Target Certificate Composition

CP validity requires policies frozen independently of the certification role and i.i.d. target-query outcomes under the stated population interpretation. For candidate and endpoint error allocations α_c, α_e , the probability that either upper bound misses its true risk is at most $\alpha_c + \alpha_e$, without requiring independence between the two bounds. Any accepted action then has risk at most δ outside that event. The original (.05, .05) allocation sums to .10; the sensitivity (.025, .025) sums to .05. This is not a .05 bound simultaneously over twenty targets, nor a bound on risk conditional on acceptance.

B Decision Details and Statistical Reconstruction

B.1 Per-Target Certificate Audit

Table 1 shows all twenty decisions. C and E mean candidate and endpoint. S/A abbreviate the two datasets; x_c, x_e count failures out of $n = 500$. $U_c(.05)$ is the original candidate bound. The stricter candidate and endpoint bounds allocate .025 each. Empirical evaluation risk uses 1,000 other queries; gain is relative mean-NDC reduction against the same target endpoint. All selected shifts are identical between the two procedures.

B.2 Crossed Bootstrap

Write D_{bq}^π and Z_{bq}^π for distance cost and binary failure on target b and shared query q , after the decision is frozen. For each of 5,000 seed-991 replicates, independently draw multinomial build weights w_b and query weights v_q , each normalized to sum one. Use the *same* query weights for all builds and all policies. Compute

$$\bar{D}^{\pi*} = \sum_{b,q} w_b v_q D_{bq}^\pi, \quad g^* = 1 - \bar{D}^{\pi*} / \bar{D}^{\text{end}*}.$$

Risk and matched old/new differences use the same product weights. Report the 2.5th and 97.5th percentiles. This resamples two crossed factors; it does not independently resample queries for each selected build. Stored source histories, chosen shifts, and certification decisions are conditioned on, not refitted. Graph-only inference instead retains the complete pair vector per query and does not resample builds.

Table 1: Twenty-target certification and completed joint-error outcomes. Bounds, evaluation risks, and gains are percentages. All endpoint bounds are at most 5%; the final action is C when $U_c(.025) \leq 5\%$, otherwise E.

Data	Seed	Shift	x_c	$U_c(.05)$	$U_c(.025)$	x_e	$U_e(.025)$	Action	Eval risk	Gain
S	1103	1	9	3.12	3.39	2	1.44	C	1.50	44.33
S	1229	1	8	2.87	3.13	5	2.32	C	1.90	44.28
S	1361	1	10	3.37	3.65	2	1.44	C	1.80	44.29
S	1499	1	6	2.35	2.59	3	1.74	C	2.00	44.51
S	1621	1	8	2.87	3.13	5	2.32	C	1.90	44.39
S	1747	1	11	3.62	3.90	2	1.44	C	1.90	44.28
S	1877	1	8	2.87	3.13	4	2.04	C	1.80	44.36
S	1999	1	8	2.87	3.13	4	2.04	C	1.70	44.35
S	2131	1	6	2.35	2.59	2	1.44	C	1.70	44.53
S	2267	1	10	3.37	3.65	2	1.44	C	1.70	44.61
A	1103	1	14	4.34	4.65	6	2.59	C	2.20	49.77
A	1229	1	17	5.06	5.39	5	2.32	E	1.30	0.00
A	1361	1	14	4.34	4.65	3	1.74	C	2.30	49.76
A	1499	1	16	4.82	5.14	4	2.04	E	0.90	0.00
A	1621	1	16	4.82	5.14	6	2.59	E	1.10	0.00
A	1747	1	15	4.58	4.90	6	2.59	C	1.50	49.84
A	1877	1	13	4.10	4.41	1	1.11	C	1.80	49.68
A	1999	1	13	4.10	4.41	5	2.32	C	2.20	49.80
A	2131	1	15	4.58	4.90	5	2.32	C	2.50	49.87
A	2267	1	16	4.82	5.14	5	2.32	E	1.00	0.00

Table 2: Mean-NDC reduction: 95% interval sensitivities (percent). Build-only conditions on queries; query-only conditions on builds. Crossed intervals are the main refresh summaries. These are not three independent replications.

Dataset	Decision	Crossed	Target-build only	Shared-query only
SIFT	Original	[42.76, 46.01]	[44.33, 44.47]	[42.86, 46.07]
SIFT	Joint	[42.76, 46.01]	[44.33, 44.47]	[42.86, 46.07]
Arxiv	Original	[34.32, 50.96]	[34.82, 49.81]	[43.27, 46.27]
Arxiv	Joint	[14.75, 45.00]	[14.92, 44.80]	[28.85, 30.84]

B.3 Fresh-Query Source-Slack Detail

Table 3 expands the two non-endpoint source-slack lanes; the main paper also reports the +2 endpoint lane. Each lane was fixed before target-evaluation access, and the rows remain separate estimands. Pair status summarizes the held-out target audit as below- δ /unresolved/above- δ according to its target-side CP interval; it is not a deployment certificate. Aggregate intervals resample 500 fresh target-query IDs with their full directed-pair vectors.

C Cost and Reproduction Contract

For each target, selection costs sum all seven registered grid searches. Certification charges the chosen candidate and the endpoint once each, sharing a search when their actions coincide for that query. Target truth costs $100,000 \times 1,000$ distance evaluations. Cold complete history adds nine old-build first-passing search profiles on all 2,000 role IDs, and $100,000 \times 2,000$ old-snapshot truth evaluations reused across those builds. Cached history treats those old acquisitions as sunk. This counts labeled information needed for execution, not target-evaluation truth used only for research assessment. Rebuilding the target graph is common; reconstructing unavailable historical graphs is outside the two scenarios.

Per-build saving is $s_b = \bar{D}_b^{\text{end}} - \bar{D}_b^{\pi}$; its break-even is A_b/s_b only for $s_b > 0$. The table’s aggregate is $\bar{A}/\bar{s}$, not the mean of individual ratios. Cost intervals resample paired (A_b, s_b) over ten targets, conditional on the finite acquisition records. At horizon N , net saving is $N\bar{s} - \bar{A}$. Full fallback gives $s_b = 0$ and never amortizes positive acquisition. This remains true even when the aggregate interval favors recovery.

The source package distinguishes three reproducibility levels: compile the paper and regenerate figures from compact records; check the statistics and certificate table from paired arrays/counts; reanalyze original native CSV responses using the external replay root. None of these commands constructs an index or launches an ANN search. Raw-search reproduction is a separate historical workflow requiring provisioned data, truth, native binaries, and indexes. The compact package does not purport to contain those large assets or establish independent replication.

Table 3: Fresh-query source-slack audit. Risk and relative mean-NDC reduction are percentages [95% query-bootstrap interval]. Pair status is below/unresolved/above $\delta = 0.05$ in the held-out target audit.

Impl.	Dataset	Lane	Risk	Reduction	Target B/U/A
hnswlib	SIFT	+0	1.487 [0.845, 2.260]	1.313 [1.307, 1.319]	543/9/0
hnswlib	Arxiv	+0	1.317 [0.600, 2.158]	0.000 [0.000, 0.000]	552/0/0
Faiss	SIFT	+0	1.297 [0.852, 1.865]	53.806 [53.758, 53.855]	478/74/0
Faiss	Arxiv	+0	1.724 [1.097, 2.474]	58.210 [58.165, 58.255]	528/24/0
hnswlib	SIFT	+1	1.400 [0.758, 2.159]	0.000 [0.000, 0.000]	552/0/0
hnswlib	Arxiv	+1	1.317 [0.600, 2.158]	0.000 [0.000, 0.000]	552/0/0
Faiss	SIFT	+1	0.183 [0.083, 0.316]	8.274 [8.266, 8.283]	552/0/0
Faiss	Arxiv	+1	0.674 [0.280, 1.211]	29.095 [29.063, 29.127]	552/0/0